Controlled Packet Capture and 802.11 Protocol Analysis Lab - WPA3-SAE vs WPA2-PSK

In this post, we will compare WPA3-SAE with WPA2-PSK using our PCAP Lab on 5 GHz. In our WPA-PSK post, we saw the AKM of 00:0F:AC:02 (PSK) in the beacon and analyzed the 4-Way Handshake. In our EAP-PEAP lab, we reviewed the AKM table, which identified 00:0F:AC:08 (SAE). Now let's put SAE to work!

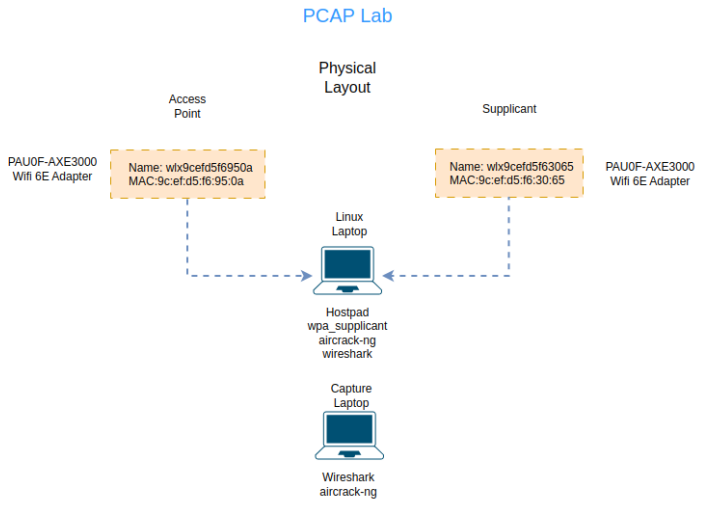
We will configure our access-point for WPA3-SAE, identify SAE in the beacon, and look at the new SAE Commit and Confirm exchange that happens before association. We will then compare a failed SAE authentication with our earlier failed WPA2-PSK authentication, so we can see exactly where each one fails. Lastly, we will take an optional look at transition mode.

Lab Setup

A picture is worth a thousand words as the saying goes! Following are diagrams depicting the Logical and Physical layout of our Lab environment. Going forward, we will reference the diagrams as required.

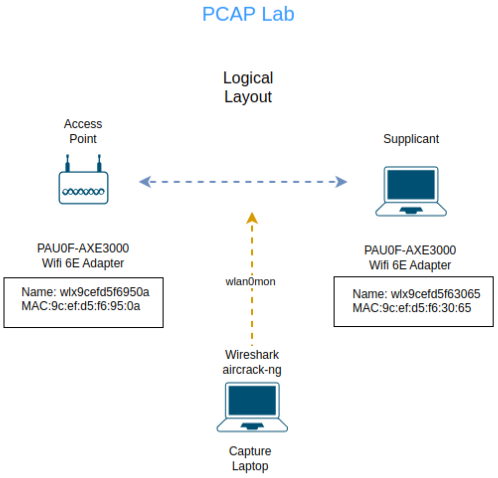
PCAP Lab Physical

Following highlights the physical components of the PCAP Lab. Please refer to the first post in this series for assistance with Lab setup if required.



PCAP Lab Logical

Logically, our lab has one supplicant (Client), one Access Point (AP), and a Laptop acting as a capture point. This laptop will follow our established process of creating a monitor interface via airmon-ng (wlan0mon) for packet capture. To minimize the length of the post, we will not be going into setup of Wireshark, as this topic has been established previously. Instead, we will review previously captured traffic to focus on analysis.



Why 5 GHz?

WPA3-SAE works on 2.4, 5, and 6 GHz. We will stay on 5 GHz, channel 36, just like our previous labs. 6 GHz requires WPA3, but it does not allow transition mode, which we explore at the end of this post. Staying on 5 GHz also means no new hardware will be required.

Before we start

- Your hostapd and wpa_supplicant builds must support SAE. Most current Linux distributions do. If hostapd rejects `wpa_key_mgmt=SAE`, your build does not include it.
- Test an SAE connection with your adapters before capturing. With mac80211 drivers, hostapd handles SAE itself, so it usually works.

Access Point

Let's kick this off by initializing our access point for WPA3-SAE. Compared to our WPA2-PSK config, only a few lines change. Notice `wpa_key_mgmt=SAE`, `sae_password`, and `ieee80211w=2`. WPA3 requires Protected Management Frames (PMF), so we set it to 2 (required). Please change the interface to align with your interface naming.

hostapd_wpa3_sae.conf

```
interface=wlx9cefd5f6950a
driver=nl80211
ssid=PCAP-Lab
hw_mode=a
channel=36
country_code=US
ieee80211n=1
wmm_enabled=1

# Beacon Interval
beacon_int=100

# WPA3-Personal (SAE)
wpa=2
wpa_key_mgmt=SAE
rsn_pairwise=CCMP
sae_password=Testlab12345

# PMF: 0 = disabled, 1 = capable (optional), 2 = required
ieee80211w=2
```

Initialize hostapd

```
sudo hostapd -i wlx9cefd5f6950a hostapd_wpa3_sae.conf
```

Result

```
wlx9cefd5f6950a: interface state UNINITIALIZED->COUNTRY_UPDATE
wlx9cefd5f6950a: interface state COUNTRY_UPDATE->ENABLED
wlx9cefd5f6950a: AP-ENABLED
```

Client (Supplicant)

Great! Now we have a functioning WPA3 access-point for our lab. Let's work on the client. We will use the following configuration file when initializing a client connection to the access-point via wpa_supplicant. Notice `key_mgmt=SAE` and `ieee80211w=2`. The client must also support PMF.

wpa3_sae_correct_password.conf

```
#ctrl_interface=DIR=/var/run/wpa_supplicant GROUP=netdev
update_config=1
country=US

network={
    ssid="PCAP-Lab"
    key_mgmt=SAE
    sae_password="Testlab12345"
    proto=RSN
    pairwise=CCMP
    group=CCMP
    ieee80211w=2

    # Send directed probe requests with this SSID
    scan_ssid=1
}
```

Initiate client connection

Prior to initiating a connection attempt to our access-point, we have started Wireshark on the capture laptop, utilizing our monitor interface wlan0mon. Perfect, let's attempt a connection.

```
sudo wpa_supplicant -i wlx9cefd5f63065 -c wpa3_sae_correct_password.conf
```

Client connecting

```
nl80211: kernel reports: Match already configured
nl80211: kernel reports: Match already configured
nl80211: kernel reports: Match already configured
wlx9cefd5f63065: SME: Trying to authenticate with 9c:ef:d5:f6:95:0a (SSID='PCAP-Lab'
freq=5180 MHz)
wlx9cefd5f63065: SME: Trying to authenticate with 9c:ef:d5:f6:95:0a (SSID='PCAP-Lab'
freq=5180 MHz)
wlx9cefd5f63065: PMKSA-CACHE-ADDED 9c:ef:d5:f6:95:0a 0
wlx9cefd5f63065: Trying to associate with 9c:ef:d5:f6:95:0a (SSID='PCAP-Lab' freq=5180 MHz)
wlx9cefd5f63065: Associated with 9c:ef:d5:f6:95:0a
wlx9cefd5f63065: CTRL-EVENT-SUBNET-STATUS-UPDATE status=0
wlx9cefd5f63065: WPA: Key negotiation completed with 9c:ef:d5:f6:95:0a [PTK=CCMP GTK=CCMP]
wlx9cefd5f63065: CTRL-EVENT-CONNECTED - Connection to 9c:ef:d5:f6:95:0a completed [id=0
id_str=]
```

hostapd - access-point

```
wlx9cefd5f6950a: STA 9c:ef:d5:f6:30:65 IEEE 802.11: associated (aid 1)
wlx9cefd5f6950a: AP-STA-CONNECTED 9c:ef:d5:f6:30:65
wlx9cefd5f6950a: STA 9c:ef:d5:f6:30:65 RADIUS: starting accounting session 68AD9183F9DDC34A
wlx9cefd5f6950a: STA 9c:ef:d5:f6:30:65 WPA: pairwise key handshake completed (RSN)
wlx9cefd5f6950a: EAPOL-4WAY-HS-COMPLETED 9c:ef:d5:f6:30:65
```

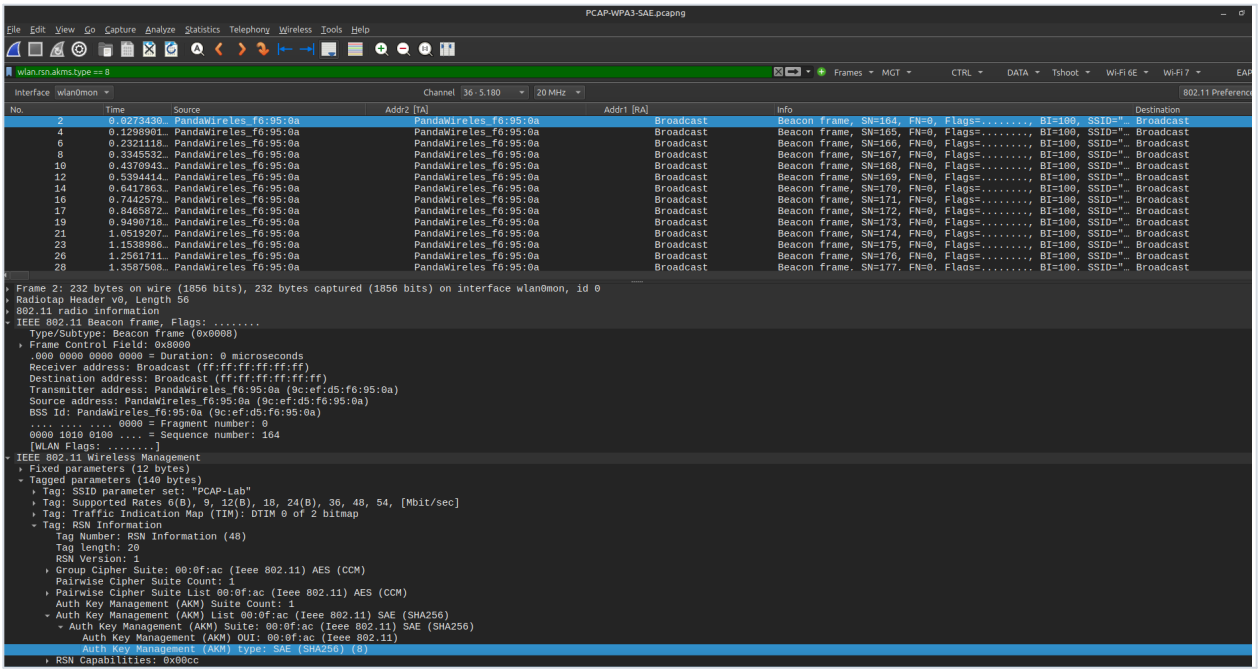
Success! Our client connected to our access-point utilizing WPA3-SAE. Let's see what this looks like via our packet capture!

Analysis

Beacon

Recall from our previous posts that Beacons carry a wealth of information. Let's start by identifying WPA3-SAE in our beacon. Apply the following display filter to show beacons advertising SAE.

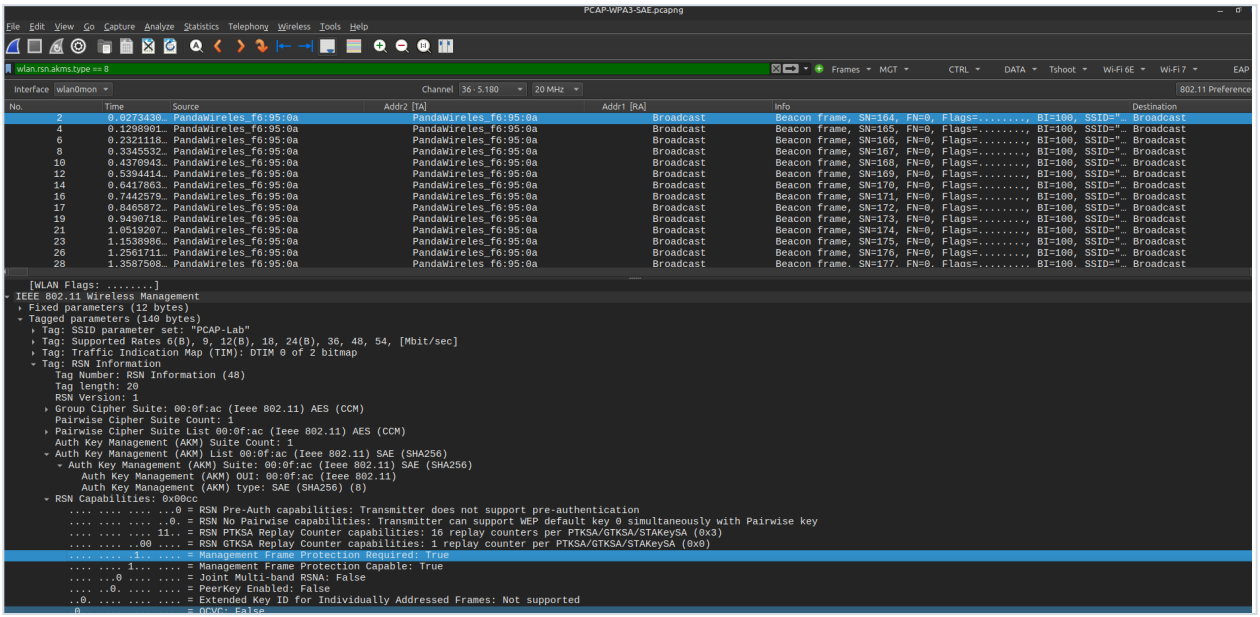
```
wlan.rsn.akms.type == 8
```



Drilling down into "Tag: RSN Information" / "Auth Key Management (AKM) List", we find the AKM suite of 00:0F:AC:08. Notice the type is now (8) for SAE, where our WPA-PSK lab showed (2) for PSK.

Next, look at "RSN Capabilities" in the same tag. Notice the two PMF bits:

- **Management Frame Protection Capable (MFPC)** - The AP supports PMF.
- **Management Frame Protection Required (MFPR)** - The AP requires PMF. Clients without PMF cannot join.



With `ieee80211w=2` set in our hostapd configuration file, both are set to True. You can filter on them directly:

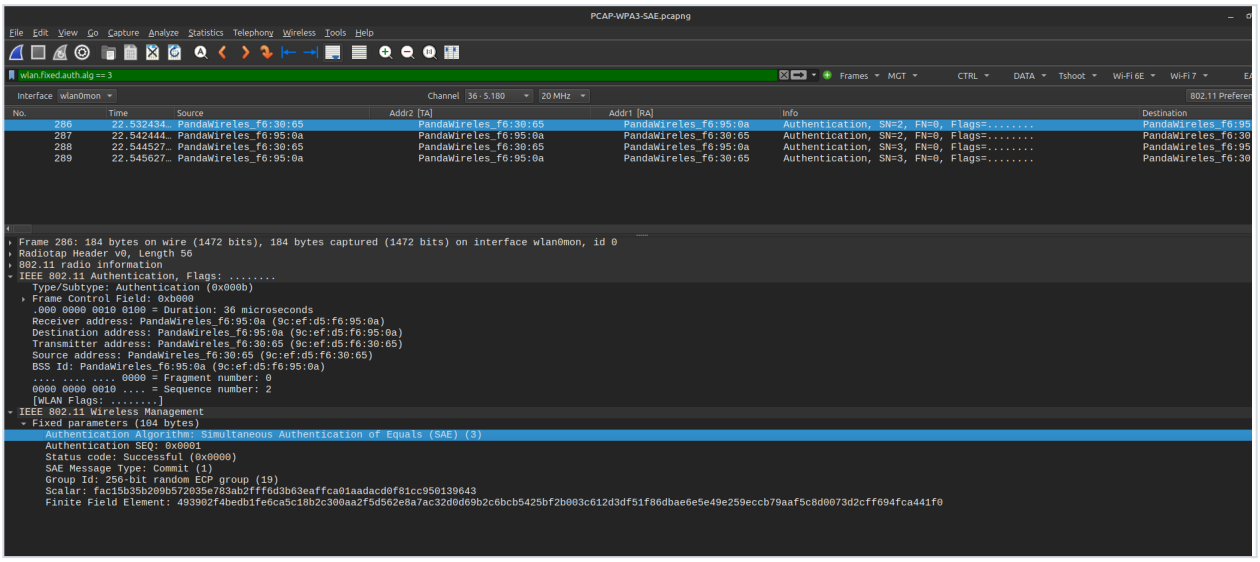
```
wlan.rsn.capabilities.mfpr == 1
```

SAE Authentication

Here is where WPA3 really differs. In our WPA-PSK lab, the Authentication frames used Open System authentication: two simple frames, and no password involved. With SAE, the password is proven during Authentication, before the client ever associates. Let's apply a filter for SAE Authentication frames.

```
wlan.fixed.auth.alg == 3
```

The Authentication Algorithm of (3) is SAE. Open System is (0).



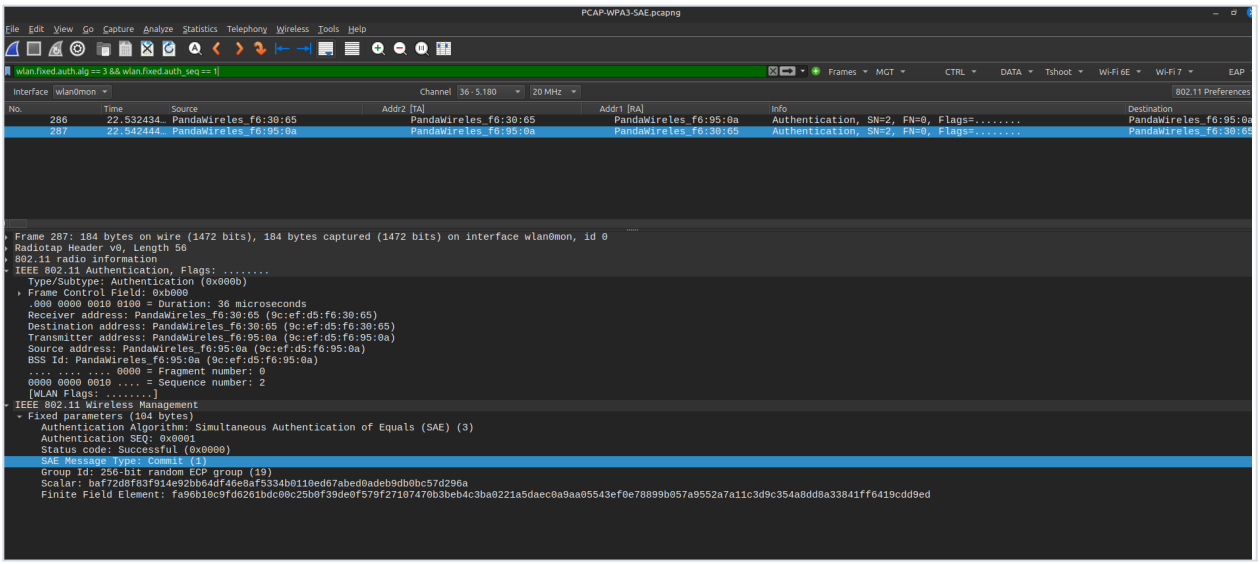
We should see four Authentication frames, two in each direction:

1. **Commit (Client to AP)** - Authentication SEQ (1). The client sends its Commit.
2. **Commit (AP to Client)** - Authentication SEQ (1). The AP sends its Commit.
3. **Confirm (Client to AP)** - Authentication SEQ (2). The client proves it derived the same key.
4. **Confirm (AP to Client)** - Authentication SEQ (2). The AP proves the same.

To look at just Commit or just Confirm frames, filter on the sequence number:

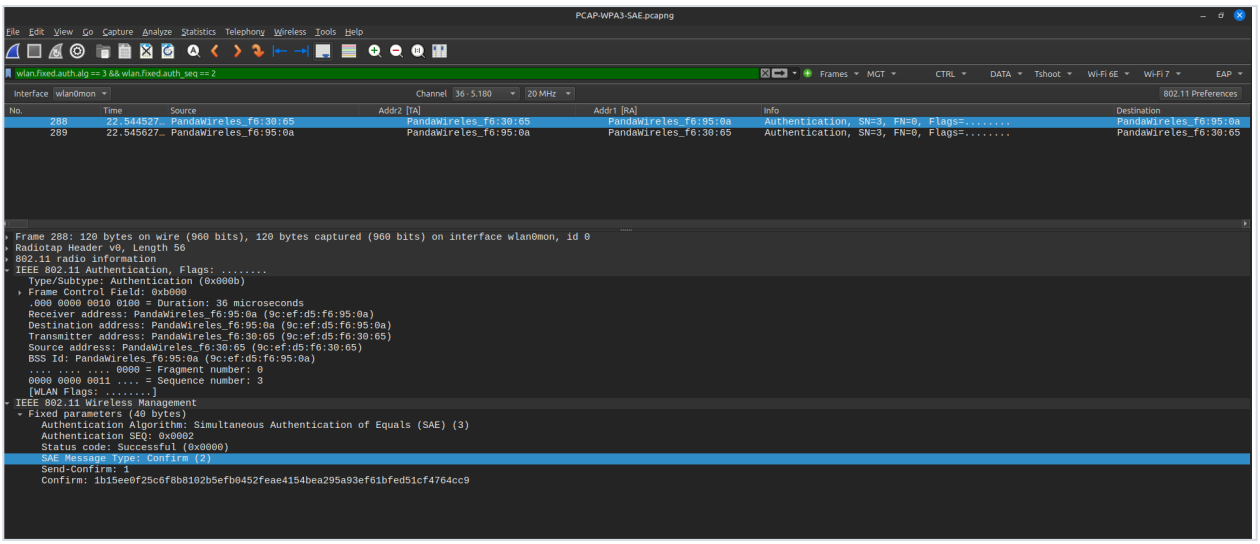
Commit Filter

```
wlan.fixed.auth.alg == 3 && wlan.fixed.auth_seq == 1
```



Confirm Filter

```
wlan.fixed.auth.alg == 3 && wlan.fixed.auth_seq == 2
```

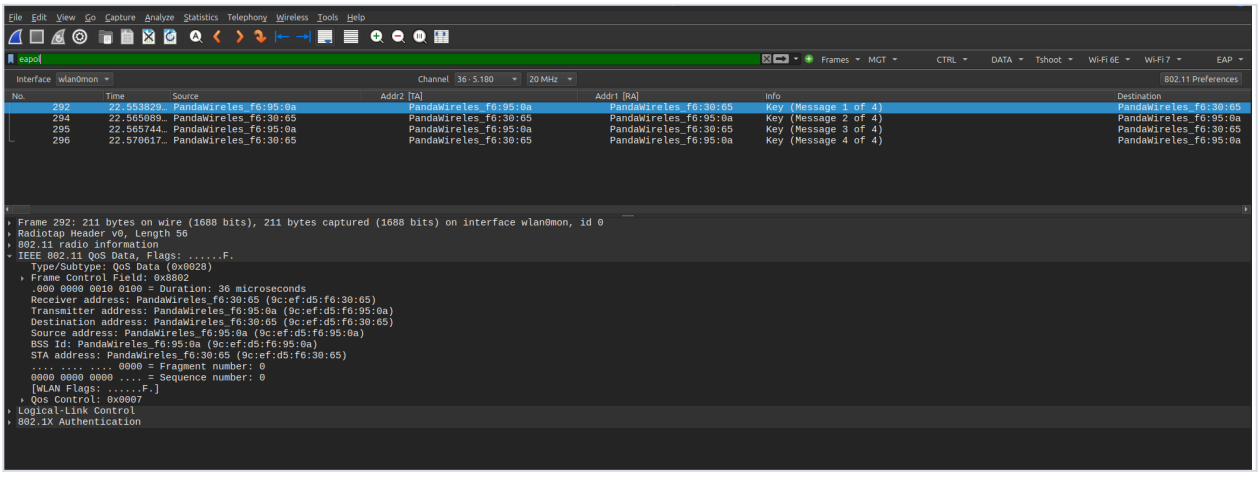


Looking inside a Commit frame, notice the "Group Id", the "Scalar", and the "Finite Field Element". The Group Id is commonly (19), a 256-bit elliptic curve group. The password itself is never sent. Both sides use these values to derive the same key.

4-Way Handshake

Once SAE Authentication completes, the client associates and the familiar 4-Way Handshake still happens. SAE does not replace it. SAE produces the PMK, and the 4-Way Handshake uses it to derive the encryption keys, just like our WPA-PSK lab.

```
eapol
```



Side by side

	WPA2-PSK	WPA3-SAE
AKM in beacon	00:0F:AC:02 (PSK)	00:0F:AC:08 (SAE)
Authentication Algorithm	Open System (0)	SAE (3)
Authentication frames	2, no password involved	4, Commit and Confirm
Password proven	4-Way Handshake, after association	SAE Confirm, before association
PMF	Optional	Required
Offline dictionary attack on a captured handshake	Possible	Not possible

WPA3-SAE Failed Authentication

Now that we have seen a successful SAE exchange, let's have a look at a failed one. For this test we will use the following wpa_supplicant configuration file. Notice we changed the `sae_password` to an incorrect password.

wpa3_sae_incorrect_password.conf

```
#ctrl_interface=DIR=/var/run/wpa_supplicant GROUP=netdev
update_config=1
country=US

network={
    ssid="PCAP-Lab"
    key_mgmt=SAE
    sae_password="WrongPassword"
    proto=RSN
    pairwise=CCMP
    group=CCMP
    ieee80211w=2

    # Send directed probe requests with this SSID
    scan_ssid=1
}
```

Initiate client connection

```
sudo wpa_supplicant -i wlx9cefd5f63065 -c wpa3_sae_incorrect_password.conf
```

Client connecting

```
wlx9cefd5f63065: SME: Trying to authenticate with 9c:ef:d5:f6:95:0a (SSID='PCAP-Lab'
freq=5180 MHz)
wlx9cefd5f63065: SME: Trying to authenticate with 9c:ef:d5:f6:95:0a (SSID='PCAP-Lab'
freq=5180 MHz)
wlx9cefd5f63065: CTRL-EVENT-AUTH-REJECT 9c:ef:d5:f6:95:0a auth_type=3 auth_transaction=2
status_code=15
^Cwlx9cefd5f63065: CTRL-EVENT-DSCP-POLICY clear_all
wlx9cefd5f63065: CTRL-EVENT-DSCP-POLICY clear_all
nl80211: deinit ifname=wlx9cefd5f63065 disabled_11b_rates=0
wlx9cefd5f63065: CTRL-EVENT-TERMINATING
```

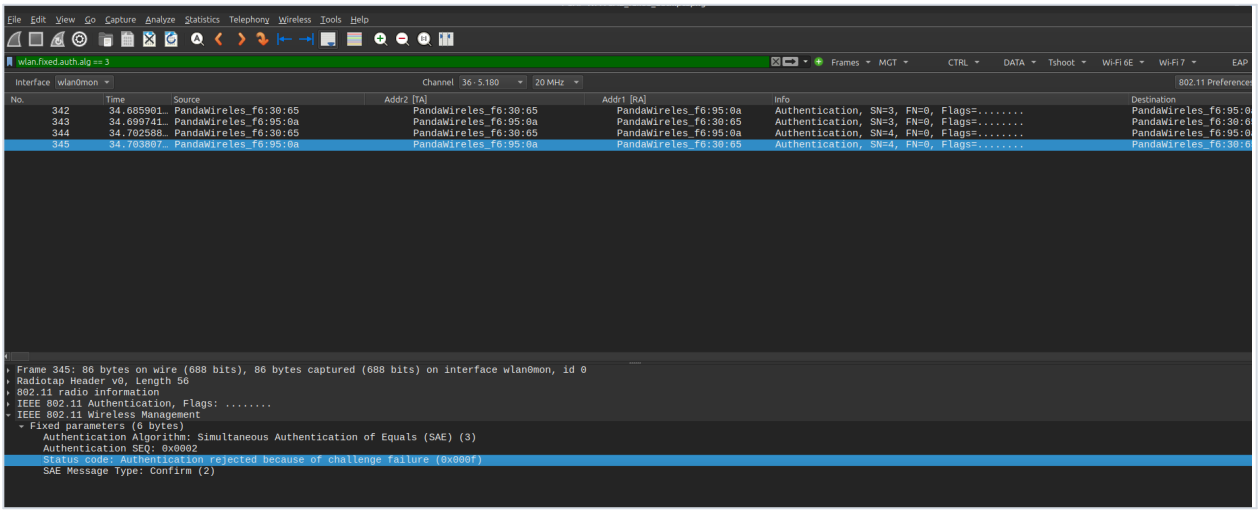
Notice `CTRL-EVENT-AUTH-REJECT`. The `auth_type=3` is SAE, `auth_transaction=2` is the Confirm, and `status_code=15` means the response to the challenge failed. In other words, the password check failed at SAE Confirm.

hostapd - access-point

```
wlx9cefd5f6950a: STA 9c:ef:d5:f6:30:65 IEEE 802.11: did not acknowledge authentication
response
```

Wireshark

```
wlan.fixed.auth.alg == 3
```



Now notice what is missing. There is no Association Request and no EAPOL. The client never gets past Authentication!!!

Where each one fails

	WPA2-PSK, wrong password	WPA3-SAE, wrong password
Authentication	Succeeds (Open System)	Fails at SAE Confirm
Association	Succeeds	Never happens
4-Way Handshake	Fails, messages 1 and 2 repeat	Never happens
What an attacker captures	A handshake they can test passwords against offline	Nothing they can test offline

This is the key difference. With WPA2-PSK, a captured handshake lets an attacker guess passwords offline, as fast as their hardware allows. With SAE, each guess requires a live exchange with the AP, so offline dictionary attacks don't work.

Optional: Transition Mode

Not every client supports WPA3. Transition mode lets an AP offer WPA2-PSK and WPA3-SAE on the same SSID, so older clients can still connect. This is allowed on 2.4 and 5 GHz, but not on 6 GHz.

hostapd_wpa3_transition.conf

```
interface=wlx9cefd5f6950a
driver=nl80211
ssid=PCAP-Lab
hw_mode=a
channel=36
country_code=US
ieee80211n=1
wmm_enabled=1

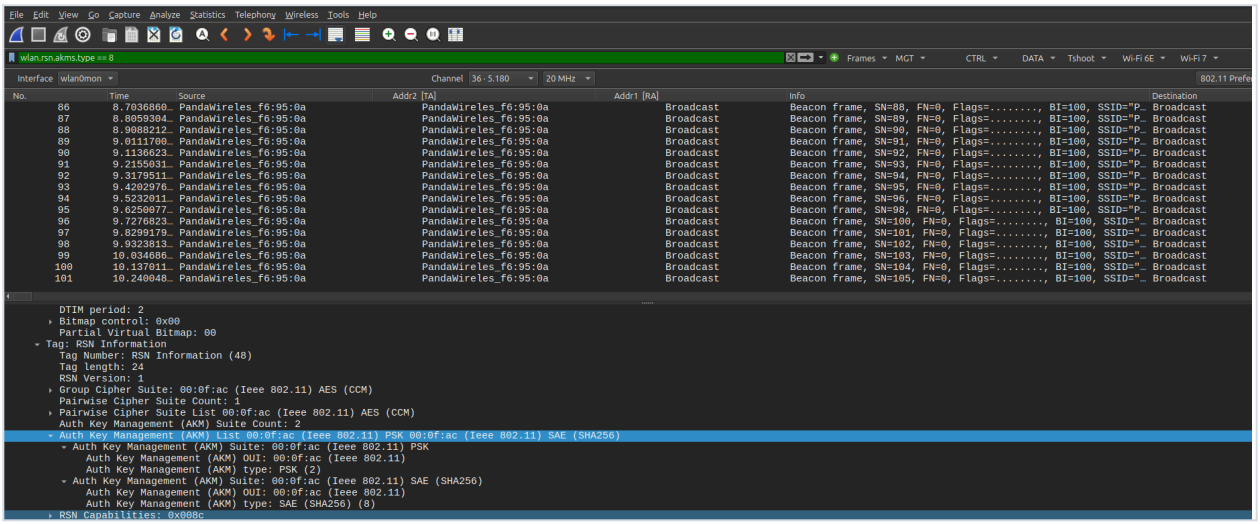
# WPA2-PSK and WPA3-SAE on the same SSID
wpa=2
wpa_key_mgmt=WPA-PSK SAE
rsn_pairwise=CCMP
wpa_passphrase=Testlab12345
sae_password=Testlab12345

# PMF capable, not required, so WPA2 clients can still join
ieee80211w=1

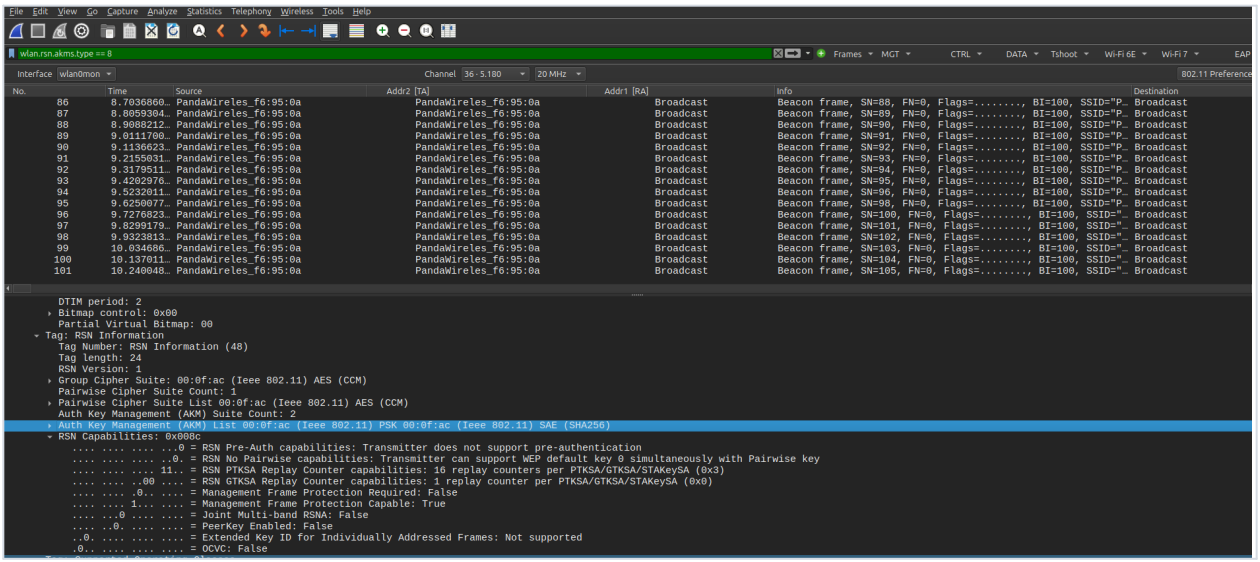
# SAE clients must still use PMF
sae_require_mfp=1
```

Initialize hostapd

```
sudo hostapd -i wlx9cefd5f6950a hostapd_wpa3_transition.conf
```



Looking at the beacon, notice the AKM List now holds two suites: 00:0F:AC:02 (PSK) and 00:0F:AC:08 (SAE). In RSN Capabilities, MFPC is True but MFPR is False, because PMF is optional for WPA2 clients.



Keep in mind, transition mode is a migration step, not a destination. An attacker could stand up a WPA2-only copy of the SSID and lure clients into connecting with WPA2-PSK, where the offline attack is possible again.

Quick Triage Tip

When a WPA3 client won't connect, the capture usually tells you where to look:

- **No SAE Authentication frames at all** - The client may not support SAE, or its config doesn't match the AP. Check the AKM in the beacon against the client config.
- **SAE Commit and Confirm repeating, no association** - Think wrong password.
- **SAE succeeds, but association is rejected** - Check PMF. A client without PMF can't join an AP that requires it. Look at the Status Code in the Association Response.

Conclusion

In this post we configured our PCAP Lab for WPA3-SAE on 5 GHz. We identified SAE in the beacon by its AKM (8) and the PMF bits in RSN Capabilities. We then saw that SAE moves password verification into the Authentication frames with the Commit and Confirm exchange, before association, while the 4-Way Handshake still follows. Comparing a failed SAE authentication with our earlier failed WPA-PSK authentication showed why SAE resists offline dictionary attacks. Lastly, we looked at transition mode and why it should be a temporary step.